

Purifying a vector state on a finite CAR stage

MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage

theorem

Every finite-stage restriction of a unit vector state can be realized by a unit vector in any irreducible CAR representation.

Statement

Let A be the completed CAR algebra: the norm completion of the increasing union of $A_n = M_{2^n}(\mathbb{C})$ under the unital embeddings $a \mapsto a \otimes I_2$ (with the fixed coordinate reindexing). Write $\iota_n : A_n \rightarrow A$ for the canonical isometric inclusion and $e_{ij}^{(n)} = \iota_n(E_{ij})$ for its matrix units. A representation is a unital complex star homomorphism into the bounded operators on a complex Hilbert space. Irreducibility means nonzero action and no proper nonzero closed reducing subspace. We use the inner product conjugate linear in its first argument and write $\varphi_{\pi,v}(a) = \langle v, \pi(a)v \rangle$ for the vector functional. Fix an irreducible representation $\pi : A \rightarrow B(H)$, any representation $\sigma : A \rightarrow B(K)$, and a unit vector $\eta \in K$. For every stage n there is a unit vector $\xi \in H$ such that $\varphi_{\pi,\xi}(\iota_n(a)) = \varphi_{\sigma,\eta}(\iota_n(a))$ for all $a \in A_n$.

Assumptions

Both H and K are complete complex Hilbert spaces. Only π is required to be irreducible with nonzero action. The representation σ is unital but may be reducible. The target vector has norm 1.

Conclusion

The two vector states agree exactly on the whole chosen matrix stage, not merely approximately on a finite list. The unit vector ξ may depend on the stage. No single vector realizing the target state on all of A is asserted.

Finite-stage purification is compatible with mixed restrictions of the target state: no purity assumption on that restriction is needed. The extra room comes from the infinite-dimensional represented root corner.

Proof route

Encode the target stage restriction as a Gram matrix, realize that matrix in the root corner of π , and reconstruct a vector by applying the first-column matrix units. Matrix-unit coefficients determine a linear functional on a full matrix algebra, and evaluation at the unit gives normalization.

▼ Proof steps

1. Set $v_i = \sigma(e_{0i}^{(n)})\eta$. The matrix-unit relations and the star property give $\langle v_i, v_j \rangle = \varphi_{\sigma,\eta}(e_{ij}^{(n)})$. Use the root-corner Gram realization theorem to choose $w_i \in \text{ran } \pi(e_{00}^{(n)})$ with the same inner products.
2. Define $\xi = \sum_i \pi(e_{i0}^{(n)})w_i$. Multiplication by a matrix unit gives $\pi(e_{pq}^{(n)})\xi = \pi(e_{p0}^{(n)})w_q$. Taking the inner product with ξ yields $\varphi_{\pi,\xi}(e_{pq}^{(n)}) = \langle w_p, w_q \rangle$.

3. The two vector functionals therefore agree on every matrix unit, and linearity extends equality to every element of A_n . At its identity, the target value is $\|\eta\|^2 = 1$ and the reconstructed value is $\|\xi\|^2$, so $\|\xi\| = 1$.

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage
- Gram realization in the root corner —
MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerFamily_with_gram
- Reconstruction from root-corner vectors —
MathlibAnnex.CStarAlgebra.CAR.reconstructedStageVector
- Matrix-unit action on the reconstructed vector —
MathlibAnnex.CStarAlgebra.CAR.representation_matrixUnit_reconstructedStageVector
- Its matrix-unit coefficients —
MathlibAnnex.CStarAlgebra.CAR.vectorFunctional_reconstructedStageVector_matrixUnit
- Matrix units determine a stage functional —
MathlibAnnex.CStarAlgebra.CAR.continuousLinearMap_eq_on_ofStage_of_eq_matrixUnit
- Equality on the whole stage —
MathlibAnnex.CStarAlgebra.CAR.reconstructedStageVector_vectorFunctional_eq_on_stage
- Normalization of the reconstructed vector —
MathlibAnnex.CStarAlgebra.CAR.norm_reconstructedStageVector_eq_one
- The infinite-dimensional corner supplying room —
MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional_range_rootCorner
- Representation and nonzero irreducibility conventions —
MathlibAnnex.Analysis.CStarAlgebra.Representation.IsIrreducible

Lean source signature (exact)

```
theorem exists_unitVector_vectorFunctional_eq_on_stage
  {H K : Type*}
  [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
  [NormedAddCommGroup K] [InnerProductSpace ℂ K] [CompleteSpace K]
  (ρ : Representation Limit H) (hp : ρ.IsIrreducible)
  (σ : Representation Limit K) (n : ℕ) (η : K) (hη : ‖η‖ = 1) :
  ∃ ξ : H, ‖ξ‖ = 1 ∧ ∀ c : Stage n,
    Representation.vectorFunctional ρ ξ (ofStage n c) =
      Representation.vectorFunctional σ η (ofStage n c)
```

In the source	Mathematical meaning
H; K; [CompleteSpace H]; [CompleteSpace K]	Both H and K are complete complex Hilbert spaces.
ρ; hp : ρ.IsIrreducible; σ	ρ is $\pi : A \rightarrow B(H)$, with nonzero irreducible action; $\sigma : A \rightarrow B(K)$ is any unital star representation, without an irreducibility hypothesis.
n : ℕ; η : K; hη : ‖η‖ = 1	Fix the stage n and the target unit vector $\eta \in K$.

In the source	Mathematical meaning
$\exists \xi : H, \ \xi\ = 1$	There is a unit vector $\xi \in H$, allowed to depend on n , with the following agreement.
$\forall c : \text{Stage } n; \text{ ofStage } n \ c$	Every matrix $c \in A_n = M_{2^n}(\mathbb{C})$, embedded as $\iota_n(c) \in A$.
<code>Representation.vectorFunctional ρ ξ (ofStage n c)</code>	The value $\varphi_{\pi, \xi}(\iota_n(c)) = \langle \xi, \pi(\iota_n(c))\xi \rangle$.
<code>Representation.vectorFunctional σ η (ofStage n c)</code>	The value $\varphi_{\sigma, \eta}(\iota_n(c)) = \langle \eta, \sigma(\iota_n(c))\eta \rangle$.
<code>... = ...</code>	These two values are exactly equal for every matrix in this one stage, using the same unit vector ξ . Equality on all of A is not asserted.

Further source notes: The proof uses `reconstructedStageVector` for the finite sum in the second step.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage`

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